

Jonathan Echavarria

CIS 344

Project #1

Subscription Box

For this project, I made a database for a subscription box service. Customers sign up for a plan and get products delivered on a regular schedule. I picked this topic because I've used these kinds of services myself, so I already had an idea of how they work.

Business Rules:

1. A customer can have many subscriptions, but each subscription belongs to one customer.
2. Each subscription is on one plan, and many subscriptions can be on the same plan.
3. A subscription can have many shipments, but each shipment belongs to one subscription.
4. A shipment can contain many products, and a product can appear in many shipments, along with the quantity of each.
5. Each product comes from one supplier, and a supplier can provide many products.
6. A subscription can have many payments, but each payment is for one subscription.

To figure out what my database needed, I researched two real subscription services, HelloFresh and Amazon Subscribe & Save. I looked at their websites and helped pages to see how customers sign up, choose a plan, get deliveries, and pay. HelloFresh charges customers automatically each week unless they skip or cancel, with the charge coming a few days before the delivery, while Amazon lets customers choose the quantity and how often they want an item

delivered, and they can pay with a credit card, debit card, or gift card. Both services allow skipping or canceling a delivery, and the products in each delivery can change, so the business needs to track what goes into each shipment and where those products come from. From this, I found the main parts my database would need to track customers, plans, subscriptions, shipments, products, suppliers, and payments. I then used these findings to write the business rules for my design.

Design:

I drew a Chen diagram by hand with seven entities: Customer, Plan, Subscription, Shipment, Product, Supplier, and Payment, with each relationship matching one of my business rules. Since one box can have many products and one product can go in many boxes, I added a Shipment_Item table to store which products are in each shipment and how many. I then made the UML diagram in MySQL Workbench, which shows all eight tables and the lines connecting them.

Challenges:

The main challenge I faced was connecting shipments and products, since one box can hold many products and the same product can show up in many boxes. On top of that, I needed a way to track how many of each product went into a box. To solve this, I added a Shipment_Item table that links each box to its products and stores the quantity of each one.

Outcome:

In the end, I built a working database for a subscription box service. It tracks customers, plans, shipments, products, suppliers, and payments. My research led to the business rules, the rules led to the diagrams, and the diagrams became the database. This project showed me how a real database is planned and built.

